

Homework 1

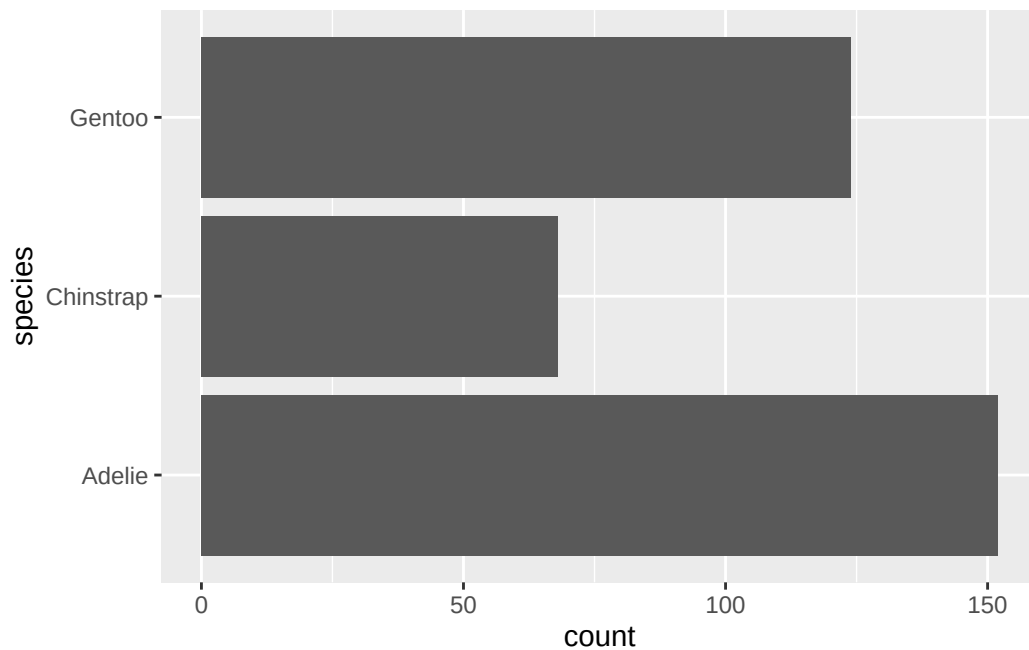
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Practice with Quarto and ggplot

1. Make a bar plot of `species` of penguins, where you assign `species` to the `y` aesthetic. How is this plot different?

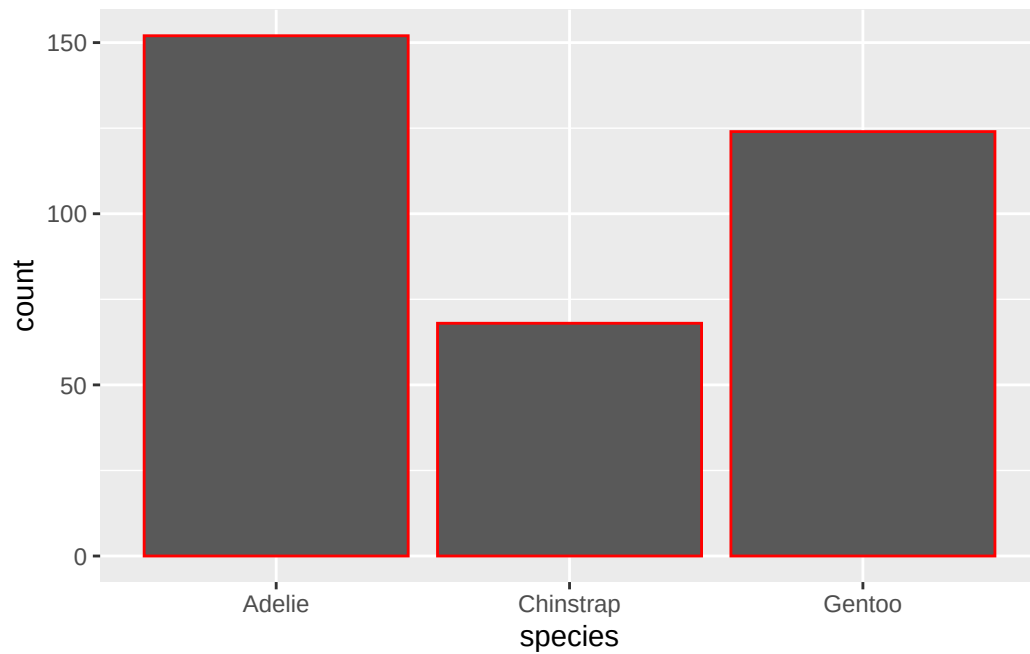
```
library(ggplot2)
library(palmerpenguins)
ggplot(penguins, aes(y=species))+ geom_bar()
```



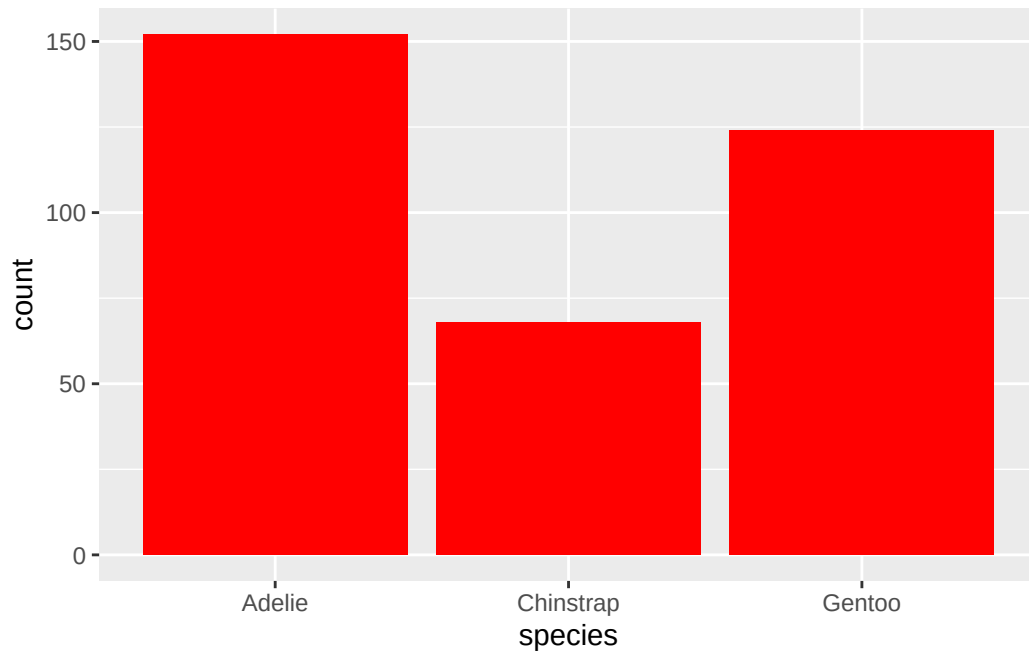
This plot is different because normally, we would put `species` on the x axis in some form of bars, here they're on the y axis.

2. How are the following two plots different? Which aesthetic, `color` or `fill`, is more useful for changing the color of bars?

```
library(ggplot2)
ggplot(penguins, aes(x = species)) +
  geom_bar(color = "red")
```



```
ggplot(penguins, aes(x = species)) +
  geom_bar(fill = "red")
```



These two plots are different because the first one only has a red colored outline for the bars and the second one completely fills the bars in with the color red. It appears that `fill` is more helpful for changing the colors of the bars.

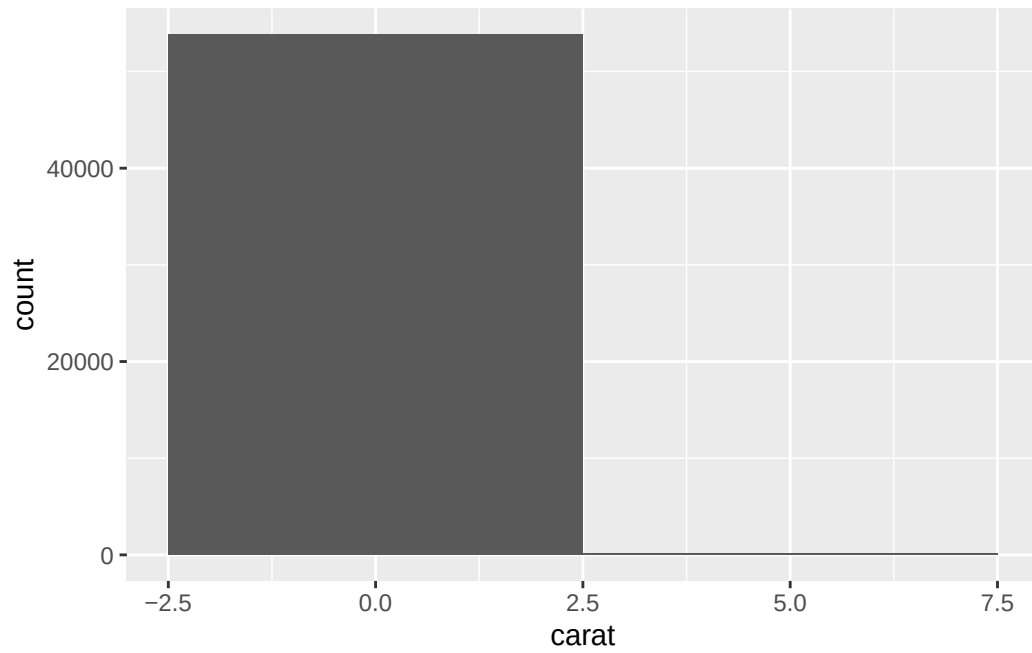
3. What does the `bins` argument in `geom_histogram()` do?

The `bins` argument controls how many bars the data is divided into.

4. Make a histogram of the `carat` variable in the `diamonds` dataset that is available when you load the tidyverse package. Experiment with different binwidths. What binwidth reveals the most interesting patterns?

```
library(ggplot2)
library(tidyverse)

ggplot(diamonds, aes(x=carat))+geom_histogram(binwidth=5)
```



The smaller the binwidth, it appears the more interesting the histogram, once the binwidth is about 5 or more, the graph just starts to resemble a large square.